

Dheepak Karan Elumalai Santhakumari

Boston, MA | (857) 339 6410 | elumalaisanthakuma.d@northeastern.edu | linkedin.com/in/dheepakkaran

SUMMARY

My engineering path began with the cooling control logic of a 210 MW generator and moved deliberately into software: a year of production backend development in Java and Python, followed by graduate work in machine learning and large language models. Electronics trained me to design within physical constraints and software engineering trained me to design for scale, which is why I work most effectively on projects that bring the two layers together.

EDUCATION

Northeastern University, Boston, MA Expected May 2028
Master of Science in Electrical and Computer Engineering
Concentration: Machine Learning, Computer Vision & Algorithms | **CGPA: 3.926**

Anna University, Tamil Nadu, India Jul 2022
Bachelor of Engineering in Electrical & Electronics Engineering
Relevant Coursework: Circuit Theory, Control Systems, EDC, Electromagnetic Fields, PE, MPMC

SKILLS

Programming: Python, C++, Java, SQL
Core CS: Data Structures, Algorithms, Computer Architecture, Database Systems
Machine Learning & AI: PyTorch, Scikit-learn, XGBoost
Data Science: Pandas, NumPy, Statistics & Probability, Linear Algebra, Data Cleaning, Web Scraping
Software Engineering: Git/GitHub, Docker, CI/CD, REST APIs, Testing, Agile, Spring Boot, FastAPI
Systems & Networks: Linux, Microservices, Apache Kafka, Networking Fundamentals, Discrete-Event Simulation
Electronics & Embedded: Circuit Analysis, PLC Programming (Ladder Logic), Sensors & Instrumentation, Industrial Control
Certifications: IBM Artificial Intelligence and Data Science Professional Certificate

WORK EXPERIENCE

Guardian Life | Software Engineer | Chennai, IN April 2022 - July 2023

- Developed backend services and REST APIs for insurance policy processing using Java Spring Boot, Python FastAPI, PostgreSQL, Apache Kafka and AWS, automating asynchronous workflows for customer-facing applications.
- Optimized database queries, indexing and concurrent request handling, improving API performance by 30%, reducing database latency by 25% and lowering application response time by 20%.
- Built secure Dockerized services with OAuth 2.0 authentication and worked with cross-functional teams to integrate machine learning models into production APIs, improving recommendation accuracy by 85%.

PROJECTS

Multilingual LLM Fine-Tuning: Tamil-English (Tanglish) Q&A Assistant May 2026
Stack: Python, PyTorch, HuggingFace (Transformers, PEFT, TRL), QLoRA, Pandas, Gradio, Google Colab (NVIDIA T4)

- Open LLMs answer Tamil-English code-switched questions poorly and often reply in Tamil script, so built the first open-source Tanglish technical Q&A dataset of 4,415 samples — 92% scraped from real user comments across 12 YouTube tech channels, the rest generated with GPT-4o-mini using structured JSON output.
- Fine-tuned Llama-3.1-8B-Instruct with QLoRA (4-bit, LoRA r=8) on a single T4 GPU, cutting GPU memory use by 75% and test perplexity by 78% (57.05 to 12.40), and released the 26 MB adapter, dataset and a Gradio demo with under 3 second rt.

FairShare-WiFi: Algorithmic Fairness Modelling in Shared Wireless Networks April 2026
Stack: Python, NumPy, Discrete-Event Simulation, Contextual Bandits (Kernel UCB), Matplotlib

- Public Wi-Fi gets unfair under congestion because user intent is not observable, so modelled an access point as a scheduling system with discrete-event simulation, generating heterogeneous traffic from stochastic models and inferring user behaviour only from measurable traffic patterns instead of content inspection.
- Implemented FIFO, Priority and Weighted Fair schedulers plus DKernelUCB, a kernel-based contextual bandit that learns allocation from traffic context, and compared all four on throughput, latency, packet drops and Jain's Fairness Index to show the fairness-versus-efficiency trade-off.

PLC-Based Stator Water Cooling Control System | NLC India Limited, Neyveli, IN March 2022
Stack: PLC Ladder Logic Programming, Flow / Pressure / Conductivity Sensors, Relay Logic, Industrial Automation

- Completed as a 2-month on-site industrial project at NLC India Limited, a Navratna Government of India power utility operating over 5,000 MW; a 210 MW lignite-fired generator trips off the grid if stator cooling water flow, pressure or conductivity stays out of limits for more than 60 seconds, and existing relay-and-timer control gave operators no continuous monitoring.
- Designed a PLC ladder-logic scheme that reads all three parameters from flow monitors and conductivity meters, automates duty/standby pump changeover on drive faults, and raises alarm and trip signals within the safety window, verified against live plant operating limits observed on site.